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Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
 B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2024
Math 2123
 (Mathematics-III)

Full Marks: 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable value for any missing data.

Section – A

1. (a) A freshly cast concrete slab has a temperature field: $T(x, y, z) = 40 - 0.1x^2 - 0.05y^2 - 0.2z$ ($^{\circ}\text{C}$) where x, y, z are in meters. Find
 - (i) the temperature gradient at $(2, 3, 0.5)$
 - (ii) the direction of maximum heat flow.
- (b) Find the constants a, b, c so that $\vec{F} = (x + 2y + az)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational and hence find the corresponding scalar potential. (12)
- (c) Find the value of the constant a and b such that the surface $ax^2 - byz = (a + 2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$. (12)
2. (a) Find the work done in moving a particle in the force field $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along
 - (i) the straight line from $(0, 0, 0)$ to $(2, 1, 3)$,
 - (ii) the curve defined by $x^2 = 4y, 3x^3 = 8z$ from $x = 0$ to $x = 2$
- (b) Evaluate $\int_S \vec{F} \cdot \hat{n} \, ds$ where S is the surface $x + y + 2z = 4$ cut by the planes $x = 0, y = 0, z = 0$ and $\vec{F} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}$. (12)
- (c) Let $\vec{F} = 2xz\vec{i} - x\vec{j} + y^2\vec{k}$. Evaluate $\iiint_V \vec{F} \, dv$, where V is the region bounded by the surfaces $x = 0, y = 0, y = 6, z = x^2, z = 9$. (11)
3. (a) Find $L\{F(t)\}$, where $F(t) = \begin{cases} 3t; & 0 < t < 2 \\ 6; & 2 < t < 4 \end{cases}$ and $F(t + 4) = F(t)$. (11)
- (b) Evaluate $\int_0^{\infty} t^3 e^{-t} \sin t \, dt$. (12)
- (c) Find the Laplace transform of $\frac{\sin at}{t}$. Does the Laplace transform of $\frac{\cos at}{t}$ exist? Justify. (12)
4. (a) State the convolution theorem. Use the convolution theorem to find:

$$L^{-1} \left\{ \frac{1}{(s+1)(s^2+1)} \right\}$$
- (b) Find the inverse transform of $\frac{s}{s^2 - 2s + 35}$. (10)
- (c) Using Laplace transform, solve $Y'' + 9Y = \cos 2t$ if $Y(0) = 1, Y\left(\frac{\pi}{2}\right) = -1$. (13)

Section – B

5. (a) Define with an example and two properties : (i) Orthogonal matrix, (ii) Elementary and (iii) Nonsingular matrix. (9)
- (b) Are vectors $[0 \ 2 \ 3 \ 4]$, $[2 \ 3 \ 0 \ 1]$, and $[4 \ 8 \ 3 \ 6]$ linearly independent? If so, why? If possible, express an independent vector as a linear combination of dependent vectors. (13)
- (c) What is meant by invertible matrix? If exists, find the inverse of the following matrix (13)
- A by elementary row transformations where, $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$
6. (a) What do you mean by the rank of a matrix? Reduce the matrix $\begin{bmatrix} 0 & 3 & -1 \\ 1 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$ to its (18)
- Row echelon form, Canonical form, and Normal form. If possible, find its inverse matrix as a by product of the process to obtain its normal form. Also determine its rank.
- (b) What is the value of K for which the following system has : (i) an unique solution, (ii) many solutions, (iii) no solution? If possible, find the unique solution of the following system using the Gauss elimination method (elementary row operations): (17)
- $$\begin{bmatrix} K & 3 & 2 \\ 0 & 6 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} K \\ 1 \\ 1 \end{pmatrix}$$
7. (a) Given $-2x + y + 3z = 0$, $2y + z = 0$, $x - 4y + Tz = 0$, What is the value of T so the system of equations has: (i) trivial solution (ii) nontrivial solution? If possible, find the nontrivial solution of the system for some value of T, and express the nontrivial solution in parametric form. (15)
- (b) Write down 2 properties for any eigensystem. Find the eigenpair and corresponding eigenspace of A, Where $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Moreover, mention algebraic multiplicity and geometric multiplicity of the above problem. Also, find the eigenvalues and corresponding eigenvectors of the matrix $A - 4I$, where I is the identity matrix. (20)
8. (a) A space curve is defined by the position vector $5 \cos t \hat{i} + 5 \sin t \hat{j} - \hat{k}$. Find the unit tangent vector $\hat{T}$, normal vector $\hat{N}$, and binormal vector $\hat{B}$. Calculate the radius of curvature, acceleration, and the tangent & normal components of acceleration. Additionally, determine the trihedral plane that contains the binormal vector $\hat{B}$ and normal vector $\hat{N}$ at $t = \frac{\pi}{2}$. (28)
- (b) A force $\vec{F} = -t \hat{i} + t^2 \hat{j}$ is acting on a moving particle with a mass of 2 units. Find its velocity such that the velocity at $t = 0$ is $5\hat{i}$. (07)

$$\hat{T} = \frac{\frac{dA}{dt}}{\text{Its value}}, \quad \hat{N} = \frac{\frac{d\hat{T}}{dt}}{\text{Value}}$$

$$\hat{B} = \hat{T} \times \hat{N}$$

$$A_t = a \cdot \hat{T} \text{ is tangential component of acceleration}$$

$$\sqrt{a^2 - A_t^2} = A_n$$

$$K = \frac{|\hat{T}|^2}{|\hat{N}|^2}$$

$$\frac{1}{K} = \rho$$